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Autonomous Indoor Rescue Drone Guided by Sound and LLMs

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Abstract

In high-risk indoor emergencies such as fires, autonomous drones can play a vital role in accelerating rescue operations while reducing risk to human responders. This paper presents a sound-guided indoor rescue drone enhanced by Large Language Model (LLM)-based multimodal reasoning to enable adaptive perception and decision-making in GPS-denied environments. The proposed system integrates sound strength estimation, vision-based scene understanding, and LLM-driven navigation to guide the drone toward a target sound source while avoiding obstacles. The approach was validated using a real DJI Tello drone in indoor corridor environments. Experimental results show that the drone successfully navigated toward the target sound source in 75% of test scenarios. On average, 6–10 LLM-generated navigation decisions were required to localize the sound source. Each perception–decision cycle required approximately 20–30 seconds, primarily due to LLM inference latency. Despite this limitation, the system demonstrated robust multimodal reasoning and safe navigation without collisions in all successful trials.

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1. Introduction

Fires in indoor environments pose a severe threat to human life and property. According to the National Fire Protection Association (NFPA), an estimated 354,400 home structure fires occur annually in the United States, causing approximately 2,620 civilian deaths, 11,220 injuries, and \$7.7 billion in direct property damage. In Saudi Arabia, fire incidents in residential and commercial buildings have been increasing, with emergency response teams facing significant challenges due to restricted visibility, complex layouts, and hazardous conditions. Traditional search-and-rescue methods often rely on human intervention, which can be delayed or hindered by environmental risks. These

limitations highlight the need for autonomous robotic systems to assist in indoor emergency scenarios. Recent advancements in artificial intelligence (AI), robotics, and autonomous navigation have paved the way for innovative solutions in disaster response. Drones, in particular, offer rapid deployment, maneuverability, and real-time data acquisition, making them ideal candidates for indoor search-and-rescue operations. However, most existing systems rely heavily on visual data, which becomes ineffective in smoke-filled environments. To address this gap, our research introduces a sound-guided indoor rescue drone powered by a Large Language Model (LLM). By integrating sound recognition for fire detection, vision-based navigation for obstacle avoidance, and LLM-driven decision-making, the proposed system enhances situational awareness and autonomous path planning. This approach aims to improve rescue efficiency in hazardous indoor environments, ultimately reducing response time and risk to human rescuers. Indoor fires present serious challenges for search-and-rescue missions due to limited visibility, structural complexity, and hazardous conditions. The remaining parts of this paper include the literature review in Section 2, the proposed method in Section 3, the results in Section 4, followed by the conclusion in Section 5.

2. Literature Review

Recent advancements in integrating sound-based guidance and Large Language Models (LLMs) into Unmanned Aerial Vehicles (UAVs) have led to significant improvements in autonomous drone operations, particularly in complex environments such as fire detection and emergency response. Yamada et al. [1] conducted a study evaluating sound source tracking methods using multiple drones, focusing on various source distances and signal-to-noise ratios. Their research highlights the importance of auditory information processing for drones in disaster site applications, emphasizing how sound recognition techniques can enhance scene analysis in low-light conditions. Zhang et al. [2] explored how UAVs can navigate using emergency sound cues, including fire alarms and distress signals. This study demonstrates the potential for drones to effectively respond to auditory signals in critical scenarios, thereby improving emergency response operations. Chen et al. [3] presented a study utilizing deep learning techniques for sound-based localization. Their research enhances UAVs' ability to identify emergency sounds, contributing to more efficient and accurate detection of critical auditory cues in disaster environments. Li et al. [4] investigated autonomous navigation, examining how drones can autonomously navigate using reinforcement learning in unfamiliar indoor spaces, which is particularly relevant to scenarios where GPS signals are unavailable or unreliable. Wang et al. [5] examined how drones combine sound and vision to navigate complex environments. Their research underscores the importance of integrating multiple sensory inputs to enhance situational awareness and decision-making capabilities in UAVs. Nguyen et al. [6] reviewed how search-and-rescue drones integrate sound, vision, and AI for effective navigation. Their findings suggest that a multimodal approach can significantly improve the efficiency and effectiveness of UAVs in search-and-rescue missions. Joshi et al. [7] introduced Neuro-LIFT, a real-time neuromorphic navigation framework implemented on a quadrotor drone. This work leverages an LLM for natural language processing to translate human speech into high-level planning commands, facilitating more intuitive human drone interactions. Lotfi et al. [8] explored leveraging large language models for map-free off-road navigation using generative AI. In this approach, a robot receives verbal instructions, and an LLM extracts landmarks and preferred terrains for constrained navigation, highlighting the potential of LLMs in enhancing autonomous navigation capabilities. Shah et al. [9] presented LM-Nav, an embodied instruction-following system that combines large pre-trained models of language, vision, and action to enable long-horizon instruction following in complex, real-world environments. This study demonstrates the feasibility of integrating LLMs with sensory inputs to improve autonomous navigation. Paredes-Vallés et al. [10] introduced a fully neuromorphic vision-to-control pipeline for controlling a freely flying drone. Their research utilizes event-based cameras and spiking neural networks for low-latency and energy-efficient perception and action, contributing to advancements in efficient autonomous drone operations. Collectively, these studies underscore the significant potential of integrating sound-based guidance and Large Language Models into UAV systems. Such integration can enhance autonomous navigation, improve emergency response capabilities, and facilitate more intuitive human-drone interactions, aligning closely with the objectives of our proposed work. Recent research highlights the importance of auditory and visual integration in UAVs for rescue missions. Yamada et al. focused on

drone-based sound source tracking, which aligns with the work on fire sound detection. Studies by Chen and Zhang further explored sound-based UAV guidance systems.

For vision processing, references include Wang et al. and Paredes-Vallés et al., who implemented multimodal and neuromorphic vision for obstacle detection and environment mapping. Gemini-based scene understanding and Neuro-LIFT models for LLM navigation (Joshi et al.) form the foundation for LLM integration. Together, these studies inform the project's multimodal AI approach.

3. Methodology

The main steps of the proposed methodology are illustrated in Figure 1. As shown, the flowchart comprises several stages that guide the drone toward the target sound signal. These steps are explained as follows:

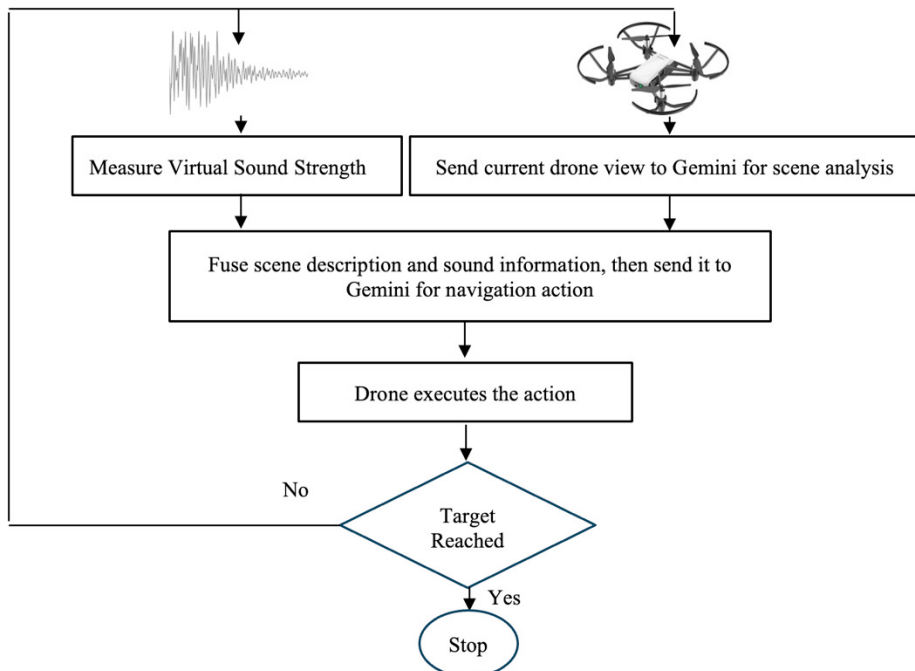


Fig. 1. The proposed approach

3.1 Measure Virtual Sound Strength:

Due to the difficulty of obtaining a real fire sound signal, this research uses a virtual sound source placed at a random location, as shown in Figure 2. Based on that location, the expected signal strength is calculated using the following formula:

$$L_2 - L_1 = -10 \log\left(\frac{r_2^2}{r_1^2}\right) \quad (1)$$

Where, L_1 and L_2 is the (dB) at positions with distances r_1, r_2 to the source

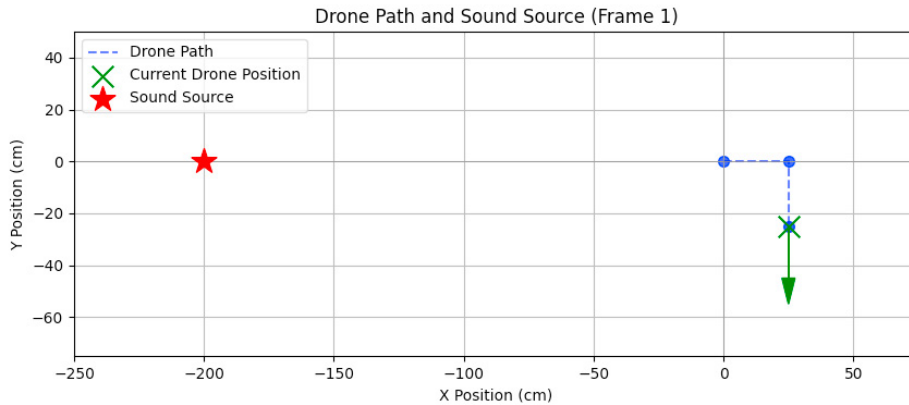


Fig. 2. Virtual sound source and drone path

3.2 Send current drone view to Gemini for scene analysis:

In this step, the drone's current camera view is sent to the Gemini vision model with the prompt below:

"Describe in detail all available objects in the scene, with their location (top left, middle, etc.), size relative to the image, estimated distance from the drone."

Then, the model will analyze the scene and provides a detailed description of all visible objects, including their approximate locations (e.g., top-left, center), relative sizes within the frame, and estimated distances from the drone.

The connection to Gemini is established through its API, as illustrated in the following code snippet:

```
# Configure Gemini API
genai.configure(api_key=GEMINI_API_KEY)
model = genai.GenerativeModel(GEMINI_MODEL_NAME)
```

3.3 Fuse scene and sound data then, send to Gemini for action output:

In this step, the estimated current and previous sound signal strengths are fused with the current scene description and sent to Gemini to generate a drone movement action, including the rotation angle and travel distance. The formulated prompt is shown below:

```
pilot_prompt = f"""
    You are an autonomous drone pilot tasked with navigating a rescue drone towards the
    source of a fire sound, while avoiding obstacles. Your decision-making is based on visual
    data, sound signal strength, current spatial context, and recent action history.
```

```
    Current Drone Spatial Context:
    {current_path_summary}
```

```
    Recent Action History:
    {recent_action_summary}
```

```
    Scene Description:
    {scene_description}
```

Instructions:

- 1- Analyze the scene description, including:
 - a- Current and previous fire sound signal strengths
 - b- Previous action and scene description
 - c- The drone's current spatial context (position, heading, distance from start)
 - d- Recent action history
- 2- Determine the optimal next action for the drone to move towards the fire sound source, explore new areas, and avoid obstacles.
- 4- If the current fire sound strength is significantly higher than the previous strength, move towards the perceived sound source.
- 5- If the current fire sound strength is significantly lower than the previous strength, rotate the drone by an angle between 90° and 180° to search in a different direction. Choose the direction of rotation based on context and obstacle avoidance:
 - a- Rotate left (positive angle) if turning left is safer.
 - b- Rotate right (negative angle) if turning right is safer.
- 7- If the scene description indicates a completely black frame (no visible objects or features), set both "rotation_angle" and "movement_steps" to 0. do not stay idle unless you are sure a collision will happen. in most cases it is better to do some exploratory movement especially if there is a change in the sound strength.
- 8- Strongly utilize the recent action history to inform decisions. Analyze the success or failure of previous moves and rotations based on changes in fire sound strength and obstacle encounters. Use this to adapt exploration strategies dynamically, favoring actions that have previously improved fire signal reception and avoiding those that led to weaker signals or collisions.
- 9- Avoid repetitive or circular behaviors. If the recent action history shows consecutive similar rotations (e.g., two or more turns in the same direction), prioritize a forward movement or a different rotation strategy to escape possible looping.
- 10- Favor forward motion when past rotations failed to improve the signal or the scene did not change meaningfully. If two or more consecutive rotations resulted in no improvement, consider switching strategy.
- 11- Track previously visited headings or directions using spatial context and avoid returning to the same heading unless sound strength justifies it.
- 12- If stuck (e.g., several moves in a row failed to increase fire strength or led to similar visual scenes), perform a large rotation (120° – 180°) and move forward to explore new space.
- 13- Use recent action outcomes and fire signal trends to guide decisions. For example:
 - If moving forward recently increased fire sound strength, favor continuing forward.
 - If actions led to weaker signals, adjust rotation significantly or change strategy.

- If multiple recent decisions failed, consider making a larger strategic adjustment (e.g., larger rotation or reversing a recent path).

14- Provide the drone's next action as a JSON object with the following keys and types:

```
"rotation_angle": [integer between {MIN_ROTATION_ANGLE} and
{MAX_ROTATION_ANGLE} degrees],
"movement_steps": [integer between {MIN_MOVEMENT_STEPS} cm and
{MAX_MOVEMENT_STEPS} cm],
"reason": "brief explanation of why this action was chosen, including how the
current and previous fire sound signal stren
```

15- when the sound strength is close to 100 dB stop the drone and descend 50cm altitude then rise 50cm and repeat that twice to indicate successful localisation of sound source

16- when the drone reaches the goal make it hover over sound source by decending 25cm then rising 25cm and repeating this decent and rise one more time. do not rotate or go any steps during this phase and reflect that in the reason as the fire is found. ""

In the remaining steps, the drone executes the generated action. If the sound signal strength becomes very close approaching 100 dB, then, the drone will stop.

Table 1. List of abbreviations.

Term	Meaning
AI	Artificial Intelligence
And another entry	Application Programming Interface
dB	Decibel
GPS	Global Positioning System
LLM	Largae Language model
UAV	Unmanned Aerial Vehicle

4. Results

The proposed rescue drone system was developed and tested in indoor environments using a real DJI Tello drone, primarily in the corridor of the SDAIA–KFUPM JRC-AI Research Center

The developed approach operates in a zero-shot manner without prior training in the target enviournment. During testing, the sound-guidance mechanism was implemented virtually through hardcoded coordinates representing simulated fire sound locations, allowing the drone to respond to directional cues toward these sources. The vision component, powered by the Gemini Vision LLM, continuously analyzed live camera input to detect and avoid obstacles such as walls and furniture, generating appropriate movement commands. For example, the drone executed the following action in response to its current visual observation, as illustrated in Figure 3.

Rotation Angle: 180

Movement Steps: 0

Action Reason:

The fire sound strength decreased after the last move. Given the close proximity of the door frame and glass panel as described in the scene, rotating 180 degrees to explore the opposite direction seems like a good strategy. The drone is very close to an obstacle, so movement steps is set to zero, while the drone rotates.



Fig. 3. Current drone view

As demonstrated in the reasoning above, the Gemini model effectively performs navigation planning by integrating the current visual context with the historical sound signal information.

The integrated reasoning process enabled the LLM to interpret both acoustic signal information and visual input, translating them into structured navigation commands communicated through the Tello SDK. The system was tested in several indoor locations, and in all cases, the drone successfully navigated toward the designated sound source. One representative case study, illustrated in Figure 4, shows the drone effectively reorienting its direction toward the target sound source while simultaneously avoiding potential collisions. These results validate the feasibility of using large language models for multimodal drone control, where sound cues guide high-level decision-making and vision-based reasoning ensures safe and adaptive indoor rescue navigation.

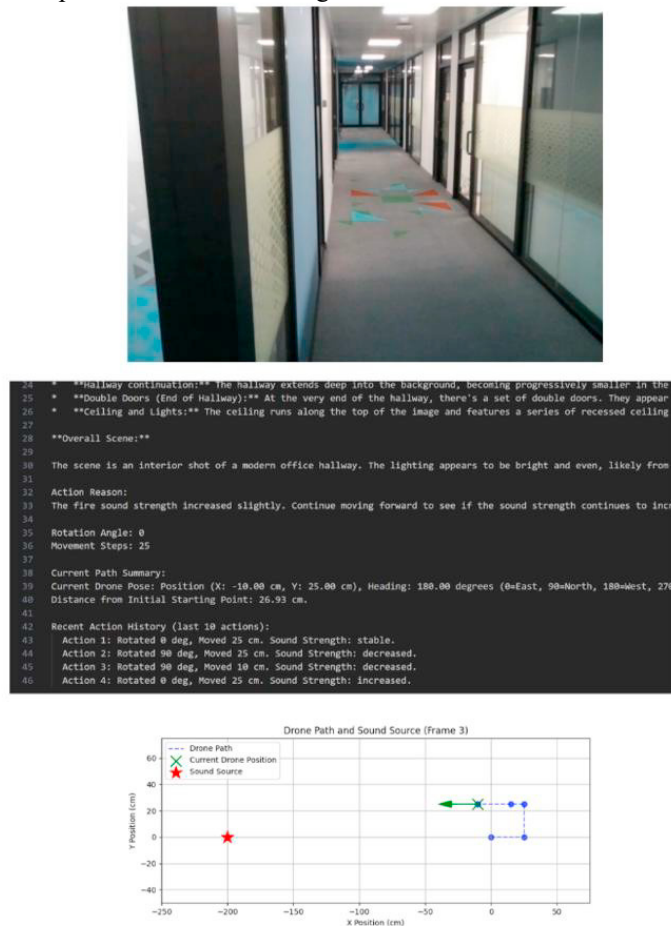


Fig. 4. Testing Tello drone in the SDAIA–KFUPM JRC-AI corridor

5. Conclusion

This work presented an autonomous indoor rescue drone guided by sound signals and controlled through Large Language Models (LLMs) with integrated vision–audio reasoning. The system demonstrated robust multimodal perception and decision-making capabilities, enabling the drone to navigate safely toward simulated fire-related sound sources while avoiding obstacles in real indoor environments. Experimental validation using a Tello drone confirmed that the LLM effectively interpreted both acoustic and visual inputs, translating them into structured navigation actions through the Tello SDK. The integration of Gemini Vision LLM for collision avoidance and sound-based directional guidance proved effective for adaptive, zero-shot indoor navigation. These results highlight the feasibility of employing LLM-driven multimodal reasoning for autonomous rescue robotics.

Nevertheless, the proposed system has several limitations. Sound guidance is currently based on a virtual sound source model, which does not capture real acoustic noise or reverberation present in fire scenarios. The use of cloud-based LLM inference introduces perception–decision latency of approximately 30–40 seconds per cycle, limiting real-time responsiveness. Additionally, experiments were conducted in structured indoor corridors with a limited number of trials, and the DJI Tello drone’s battery life of approximately 5 minutes constrains mission duration. Future work will focus on integrating real acoustic sensing, reducing latency through lightweight or onboard LLMs, and extending experimental validation to more complex indoor environments. Future work will focus on deploying lightweight, energy-efficient LLMs directly onboard drones to enhance autonomy and incorporating thermal and depth sensing for improved hazard and obstacles detection.

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